

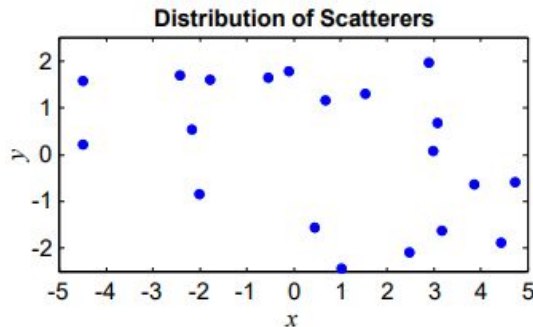
Radar Signal Models, part 2: Statistical Models of Target RCS, Clutter RCS, and Noise

EE417 / EE517
CUA Spring 2024

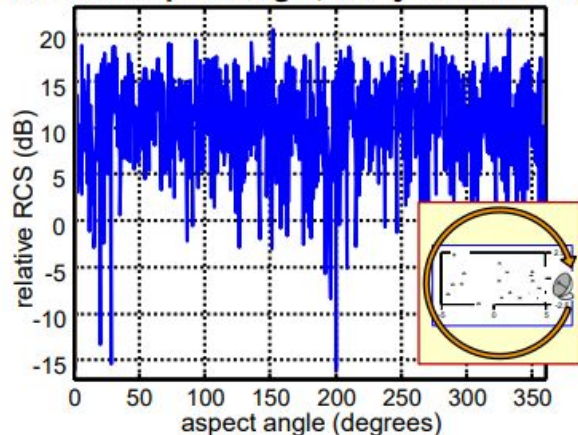
Tuesdays 7:00-9:30pm
Instructor: Prof. Kevin Russo

RCS Variation with Aspect Angle or RF

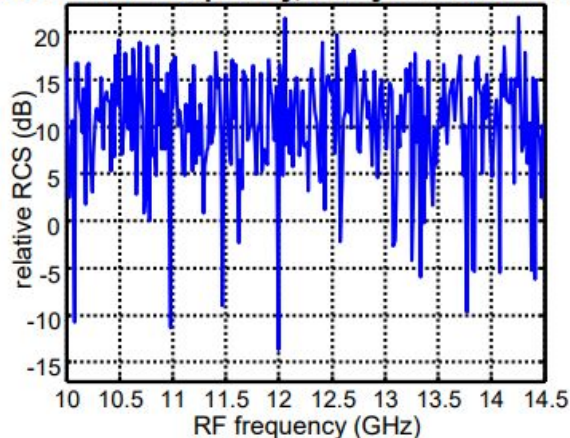
- For a target composed of many unresolved scatterers, variation in aspect angle or RF changes the phases $-(4\pi/\lambda)R_i \rightarrow$ changes RCS
- Example: a 10×5 m “target” consists of 20 unresolved scatterers
- RCS variations > 30 dB observed
- RCS very sensitive to RF or angle



RCS vs. Aspect Angle, Many-Scatterer Case

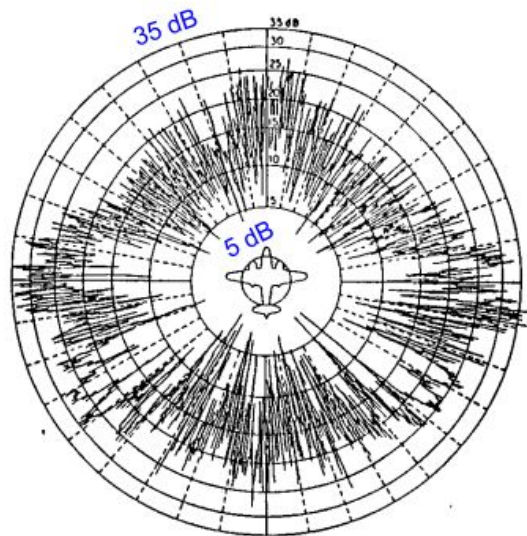


RCS vs. RF Frequency, Many-Scatterer Case



Modeling RCS Fluctuations of Complex Targets

$\lambda = 10$ cm (3 GHz)



*Measured RCS
of a B-26
“Marauder”
bomber as a
function of
azimuth angle*

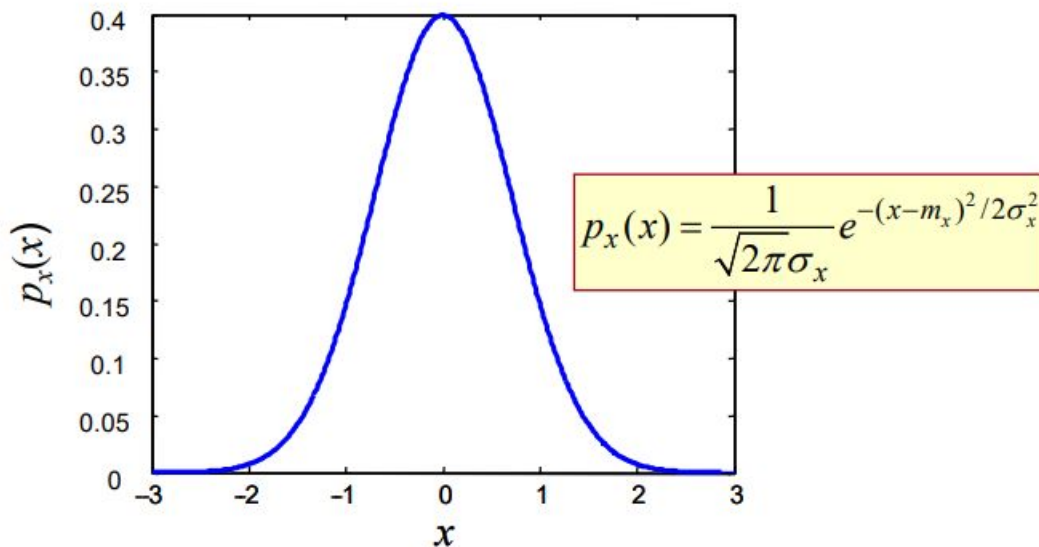


- The radar return from a typical “real world” target is the result of multiple scattering processes
 - As aspect angle changes, the number, nature, orientation, range, and phase of the unresolved scatterers contributing to the return changes
- RCS varies (“**fluctuates**”) with aspect angle or RF in a very complex manner
- RCS can vary 30+ dB
- RCS variation can be 10-15 dB over a small fraction of a degree or a few MHz
- The RCS of a complex target is generally treated as random
 - Not really, but too complicated to model any other way

Some Reminders from Probability and Random Variables

Probability Density Function

- A probability density function (PDF) $p_x(x)$ describes the values a random variable x can take on and their relative likelihood
- Example PDF: Gaussian (a.k.a. normal) with zero mean ($m_x = 0$) and unit variance ($\sigma_x^2 = 1$)
 - In this example, a measured value around zero is more likely to occur than a measured value around, say, +3 or -2



Probability of a Threshold Crossing

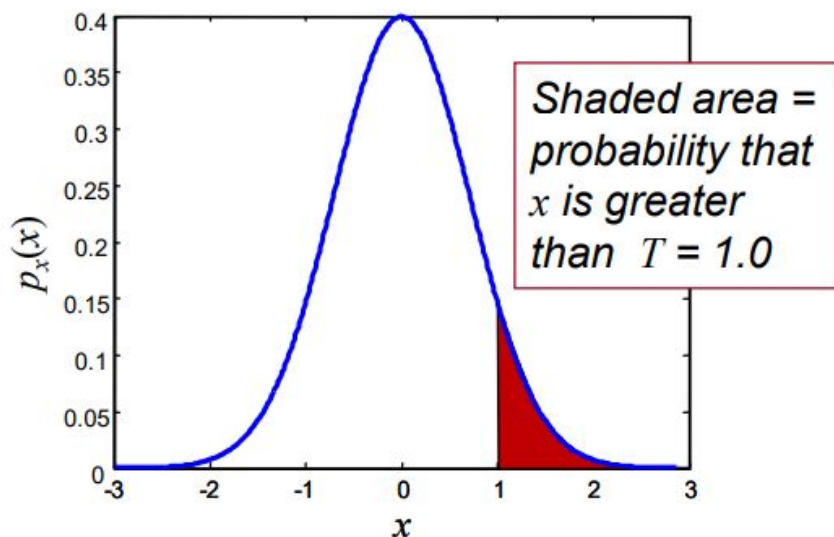
- A PDF is used to compute the probability that a random variable falls in some interval:

$$\text{Probability that } x_1 \leq x \leq x_2 = \int_{x_1}^{x_2} p_x(x) dx$$

- The probability that x exceeds a threshold T is the “right-tail” probability that it is between T and $+\infty$:

Probability that $x > T =$

$$\int_T^{\infty} p_x(x) dx$$



Expectations (or Moments) of Random Variables

- **Mean (expected value):**

$$E\{x\} = \int_{-\infty}^{\infty} x \cdot p_x(x) dx = m_x = \bar{x}$$

- **Mean-squared (average power)**

$$E\{x \cdot x^*\} = E\{|x|^2\} = \int_{-\infty}^{\infty} |x|^2 p_x(x) dx = \overline{x^2}$$

- **Variance**

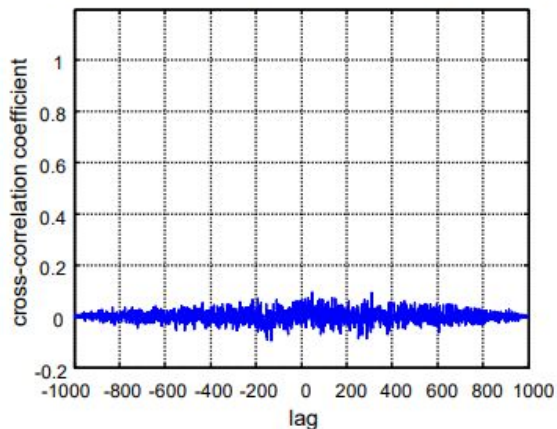
$$\begin{aligned} \text{var}\{x\} &= E\{(x - m_x)(x - m_x)^*\} = \sigma_x^2 \\ &= E\{x \cdot x^*\} - |m_x|^2 = \overline{x^2} - |m_x|^2 \end{aligned}$$

- **Note that if the mean is zero, then the variance equals the average power**

The (Cross-)Correlation Function of Two Random Signals

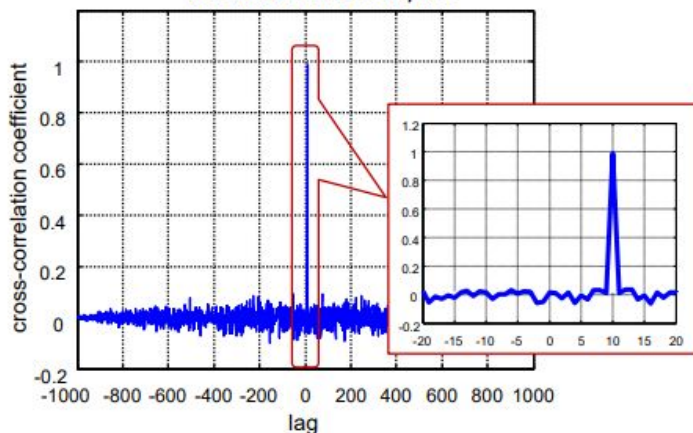
- A measure of similarity between two random signals $x[n]$ and $y[n]$, when one is offset from the other by a “lag” of m samples
- **Definition:** $\phi_{xy}[m] = E \left\{ x[n+m]y^*[n] \right\}$
- **Simulation examples:**

Normalized cross-correlation of two independent zero mean, unit variance Gaussian sequences



Low correlation for any lag shows that these two sequences are not similar to one another

Normalized cross-correlation of a zero mean, unit variance Gaussian with an identical signal except for a shift of 10 samples



High correlation peak at lag 10 shows that these two sequences are very similar, with a 10-sample offset

The Autocorrelation Function of a Random Signal

- **Cross-correlation of a signal with itself:**

$$\phi_{xx}[m] = E \left\{ x[n+m]x^*[n] \right\}$$

- **Two properties:**

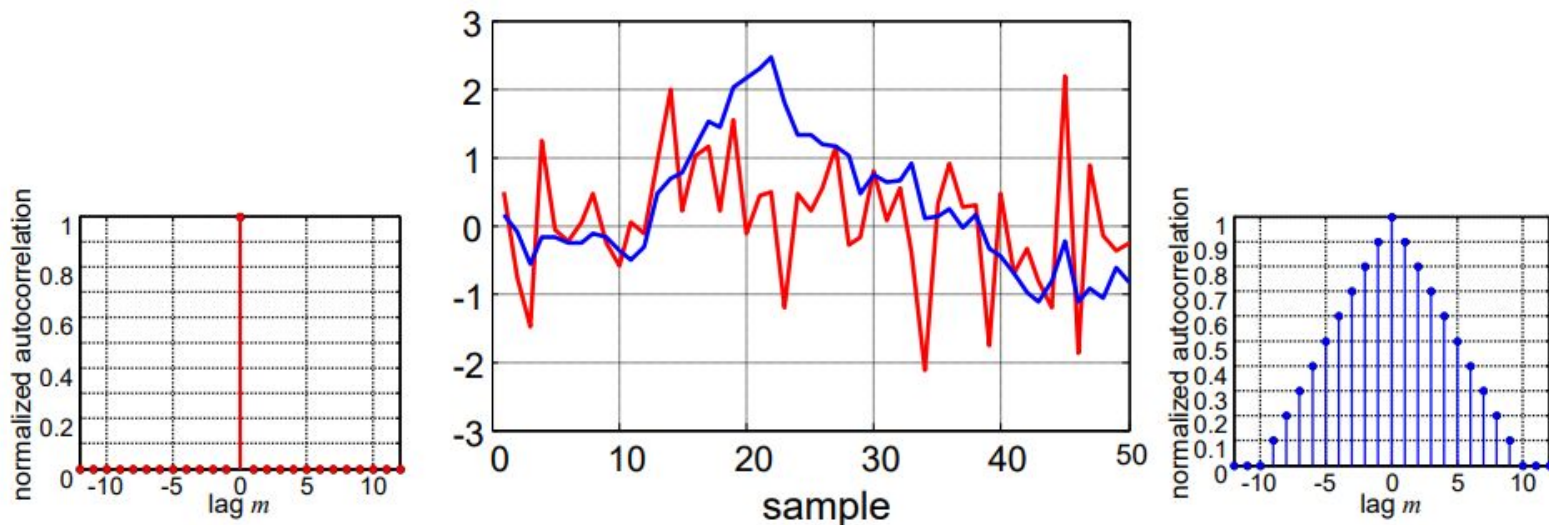
- **Average power:** $\phi_{xx}[0] = E \left\{ |x[n]|^2 \right\} = \text{mean-square}$

- **Shape:** $|\phi_{xx}[m]| \leq \phi_{xx}[0]; \quad \lim_{m \rightarrow \infty} \phi_{xx}[m] = |m_x|^2$

- $\phi_{xx}[m]$ tells us the expected degree of similarity between the current value of x and its value m samples later
- **High autocorrelation values over larger ranges of the lag m imply a more smoothly-varying process**
 - Which is equivalent to a more lowpass power spectrum

Autocorrelation Example

- **Example: two Gaussian sequences with the same mean and variance**
same mean and variance



autocorrelation of **red** data:

$\phi_{xx}[m] = \delta[m]$ (white noise,
uncorrelated sample-to-sample)

autocorrelation of **blue** data:

$\phi_{xx}[m] = 1 - m/10$ (triangle function,
correlated sample-to-sample,
correlation declines for longer lags)

The Power Spectrum

- The **power spectrum** (or **power spectral density, PSD**) of a random process is the **Fourier transform of the autocorrelation function**:

$$S_{xx}(\omega) = \mathcal{F}\{\phi_{xx}[m]\}$$

- By Parseval's theorem,

$$E\{|x|^2\} = \phi_{xx}[0] = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{xx}(\omega) d\omega$$

- So we can also get the average power from the integral (area of) the power spectrum

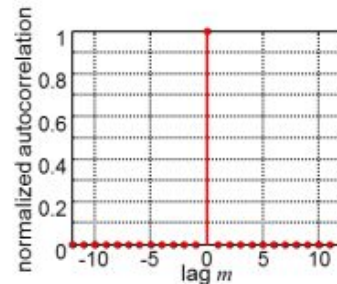
White Noise

- Consider a zero mean signal with autocorrelation

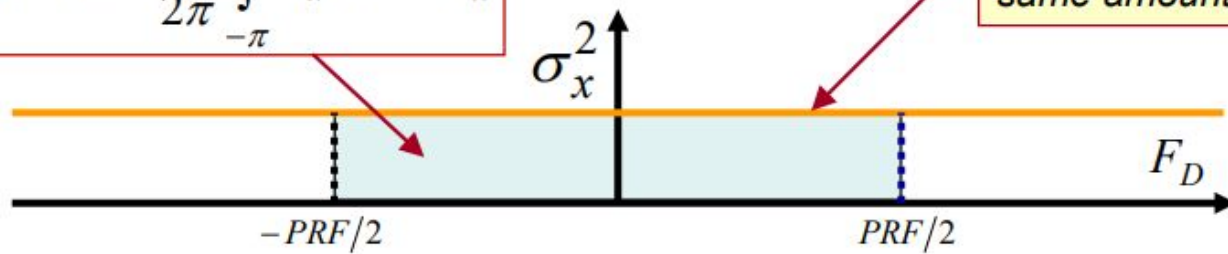
$$\phi_{xx}[m] = \sigma_x^2 \delta[m]$$

- The power spectrum of this signal is

$$S_{xx}(\omega) = \sigma_x^2, \quad |\omega| \leq \pi$$



$$\text{ave. power} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \sigma_x^2 d\omega = \sigma_x^2$$



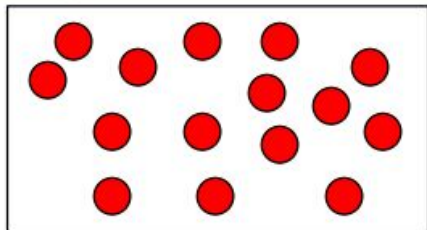
All frequencies present in the same amount.

- White noise is a very common receiver noise model

Common Statistical RCS Models

“Many Scatterer” Fluctuating Target Model

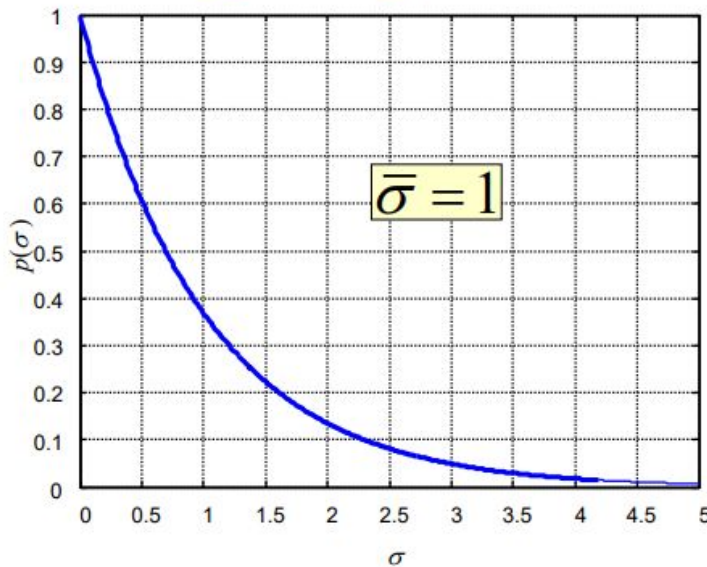
Target Modeled as a Collection
of $\approx$ Equal RCS Scatterers



This fluctuation model follows
from the Central Limit Theorem

Swerling Cases 1 and 2

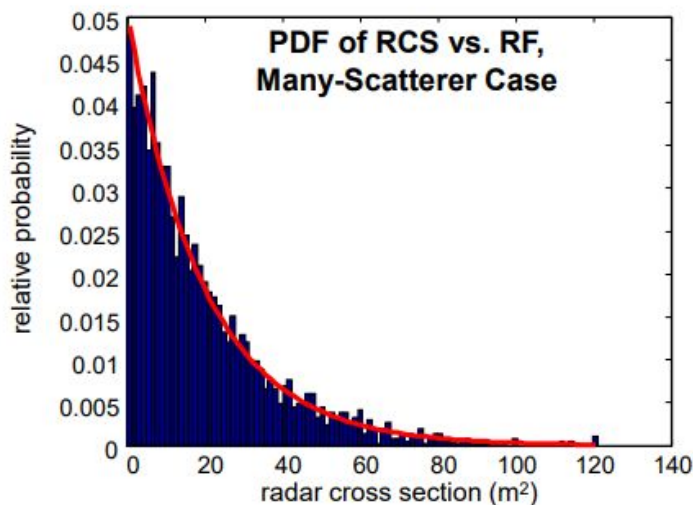
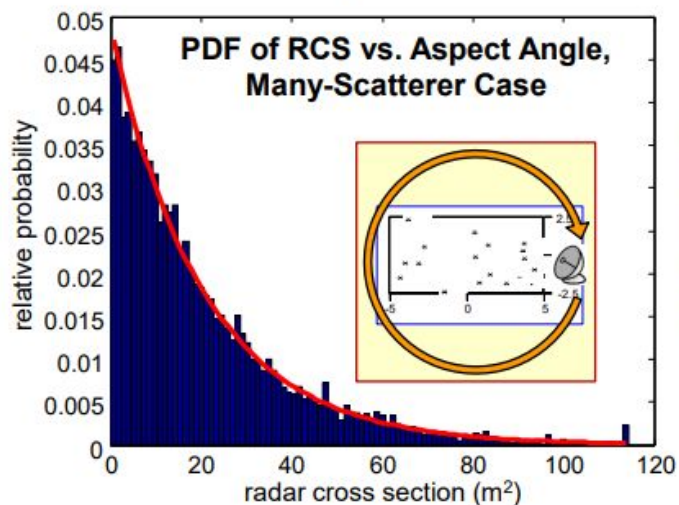
$$p_{\sigma}(\sigma) = \frac{1}{\bar{\sigma}} e^{-\sigma/\bar{\sigma}}, \quad \sigma \geq 0$$



- **Exponential RCS PDF**
 - Central chi-square of degree 2
 - Sometimes called a Rayleigh target because the corresponding voltage (not power) PDF is the Rayleigh PDF

Example Simulated vs. Theoretical RCS PDF

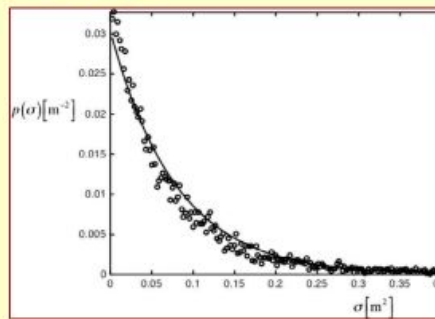
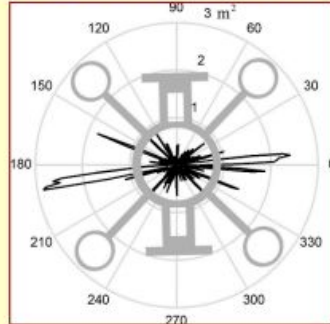
- Average histogram of linear-scale (not dB) data from ten 20-scatterer targets
- The red curve is an exponential PDF that matches the mean of the data
- The exponential PDF models the observed RCS variation with either aspect or RF well



Recent Example: 8-12 GHz Quadcopter RCS

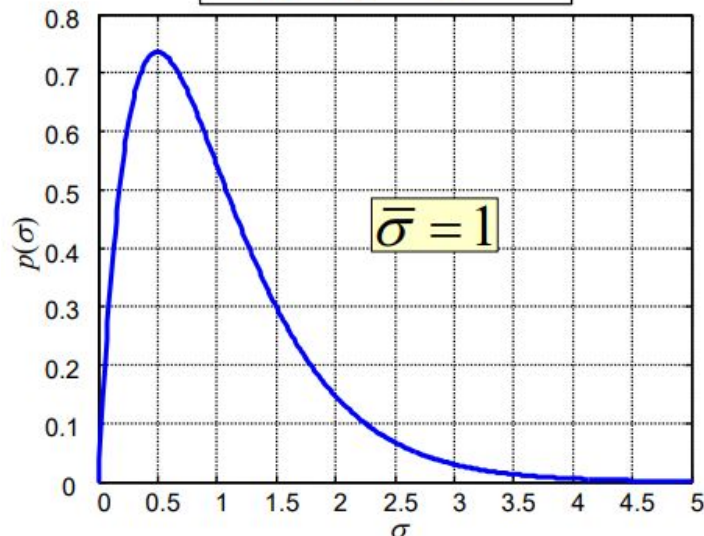
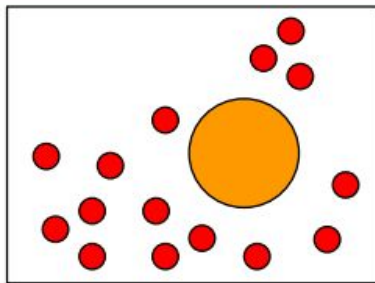
- Mean RCS (vertical polarization): -5.7 dBsm
 - Horizontal nearly the same
- RCS variation with aspect ~ 10 dB
- Exponential PDF a good fit to angle/frequency data

Pro: AirVision NT4Contra



“Dominant + Many” Fluctuating Target Model

Target Modeled as a Collection of Scatterers with One Dominant Scatterer



- The correct RCS PDF for this case is a noncentral chi-squared with 2 degrees of freedom (square root of RCS – Rician PDF)
- Swerling's approximation shown here is a central chi-squared of degree 4:

Swerling Cases 3 and 4

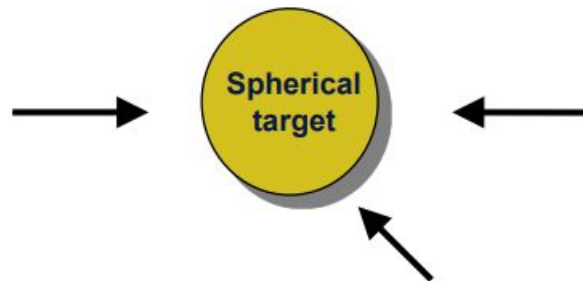
$$p_{\sigma}(\sigma) = \frac{4\sigma}{\bar{\sigma}^2} e^{-2\sigma/\bar{\sigma}}, \quad \sigma \geq 0$$

Nonfluctuating Target

- The RCS is independent of aspect angle and RF, so it is always the same value σ_0

$$p_{\sigma}(\sigma) = \delta_D(\sigma - \sigma_0)$$

- δ_D = Dirac impulse function
 - Sometimes called the “Swerling 0” or “Swerling 5” case
 - RF independence requires the sphere diameter to be “much larger than” the wavelength
- Data from a complex target can be nonfluctuating if collected with no geometry or frequency changes



Radar Calibration Spheres shown with optional polished finish

Some Common Probability Density Functions for RCS (1 of 2)

Model name	PDF for RCS σ	Comment
<i>One-parameter PDFs</i>		
Nonfluctuating, Marcum, Swerling 0, or Swerling 5	$p_{\sigma}(\sigma) = \delta_D(\sigma - \bar{\sigma})$ $\text{var}(\sigma) = 0$	Constant echo power, e.g., calibration sphere or perfectly stationary target with no radar or target motion.
Exponential, Chi-square of degree 2	$p_{\sigma}(\sigma) = \frac{1}{\bar{\sigma}} \exp\left[\frac{-\sigma}{\bar{\sigma}}\right]$ $\text{var}(\sigma) = \bar{\sigma}^2$	Many scatterers, randomly distributed, none dominant. Used in Swerling case 1 and 2 models.
Chi-square of degree 4	$p_{\sigma}(\sigma) = \frac{4\sigma}{\bar{\sigma}^2} \exp\left[\frac{-2\sigma}{\bar{\sigma}}\right]$ $\text{var}(\sigma) = \bar{\sigma}^2 / 2$	Approximation to case of many small scatterers + one dominant. Used in Swerling case 3 and 4 models.

Some Common Probability Density Functions for RCS (2 of 2)

Model name	PDF for RCS σ	Comment
<i>Two-parameter PDFs</i>		
Chi-square of degree $2m$, Weinstock	$p_{\sigma}(\sigma) = \frac{m}{\Gamma(m)\bar{\sigma}} \left[\frac{m\sigma}{\bar{\sigma}} \right]^{m-1} \exp \left[-\frac{m\sigma}{\bar{\sigma}} \right]$ $\text{var}(\sigma) = \bar{\sigma}^2 / m$	Generalization of the two preceding cases. Weinstock cases correspond to $0.6 \leq 2m \leq 4$. Higher degrees correspond to presence of a more dominant single scatterer.
Noncentral chi-square of degree 2	$p_{\sigma}(\sigma) = \frac{1}{\bar{\sigma}} (1 + a^2) \exp \left[-a^2 - \frac{\sigma}{\bar{\sigma}} (1 + a^2) \right]$ $\times I_0 \left[2a\sqrt{(1 + a^2)(\sigma / \bar{\sigma})} \right]$ $\text{var}(\sigma) = \frac{(1 + 2a^2)}{(1 + a^2)^2} \bar{\sigma}^2$	Exact solution for one dominant scatterer plus many small ones. Corresponds to Rice amplitude PDF. Ratio of dominant RCS to sum of small RCS is a^2 . $I_0(\cdot)$ is the modified Bessel function of the first kind and order zero.
Weibull	$p_{\sigma}(\sigma) = CB\sigma^{C-1} \exp [-B\sigma^C]$ $\bar{\sigma} = \Gamma(1 + 1/C) B^{-1/C}$ $\text{var}(\sigma) = B^{-2/C} [\Gamma(1 + 2/C) - \Gamma^2(1 + 1/C)]$	Empirical fit to many measured target and clutter distributions. Can have longer "tail" than previous cases. Not readily expressible in terms of $\bar{\sigma}$.
Log-normal	$p_{\sigma}(\sigma) = \frac{1}{\sqrt{2\pi s\sigma}} \exp [-\ln^2(\sigma / \sigma_m) / 2s^2]$ $\bar{\sigma} = \sigma_m \exp(s^2 / 2)$ $\text{var}(\sigma) = \sigma_m^2 \exp(s^2) [\exp(s^2) - 1]$ $= \bar{\sigma}^2 [\exp(s^2) - 1]$	Empirical fit to many measured target and clutter distributions. "Tail" is longest of previous cases. σ_m is the median value of σ . Not readily expressible in terms of $\bar{\sigma}$.

One- and Two-Parameter PDFs

- The exponential, chi-square, and impulse PDFs used in the Swerling and nonfluctuating models have only one free parameter, the mean RCS $\bar{\sigma}$ (σ_0 for the nonfluctuating case):

Swerling 1 and 2

$$p_{\sigma}(\sigma) = \frac{1}{\bar{\sigma}} e^{-\sigma/\bar{\sigma}}$$

central chi-square
of degree 2

Swerling 3 and 4

$$p_{\sigma}(\sigma) = \frac{4\sigma}{\bar{\sigma}^2} e^{-2\sigma/\bar{\sigma}}$$

central chi-square
of degree 4

Nonfluctuating

$$p_{\sigma}(\sigma) = \delta_D(\sigma - \sigma_0)$$

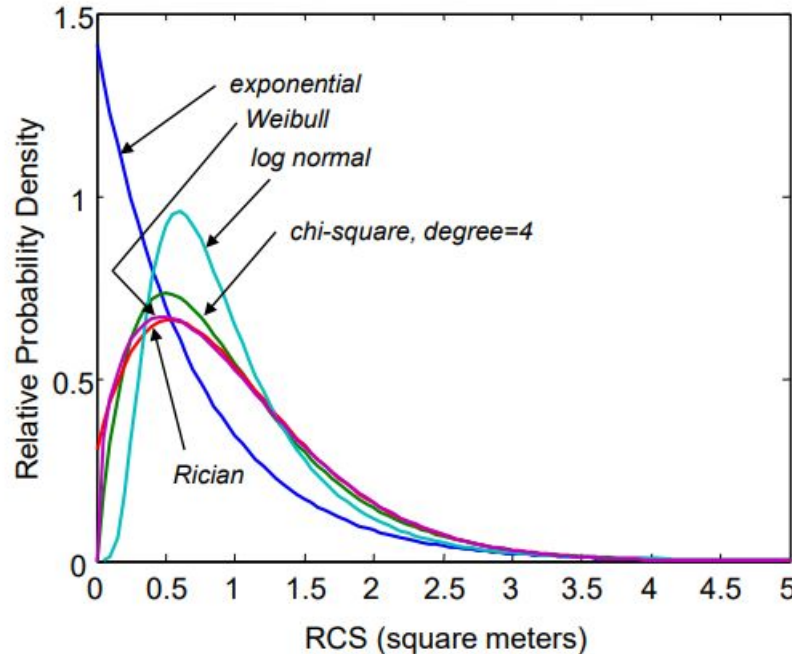
- The Rice, Weibull, Log-Normal, K distribution, and Weinstock all have two parameters
- Example: the Rice (non-central chi-square) PDF uses the mean RCS and a dominant-to-small scatterer ratio a^2 :

$$p_{\sigma}(\sigma) = \frac{1}{\bar{\sigma}} (1 + a^2) \exp \left[-a^2 - \frac{\sigma}{\bar{\sigma}} (1 + a^2) \right] I_0 \left[2a \sqrt{(1 + a^2) (\sigma / \bar{\sigma})} \right]$$

- Two parameters allows fitting to a wider range of measured data**

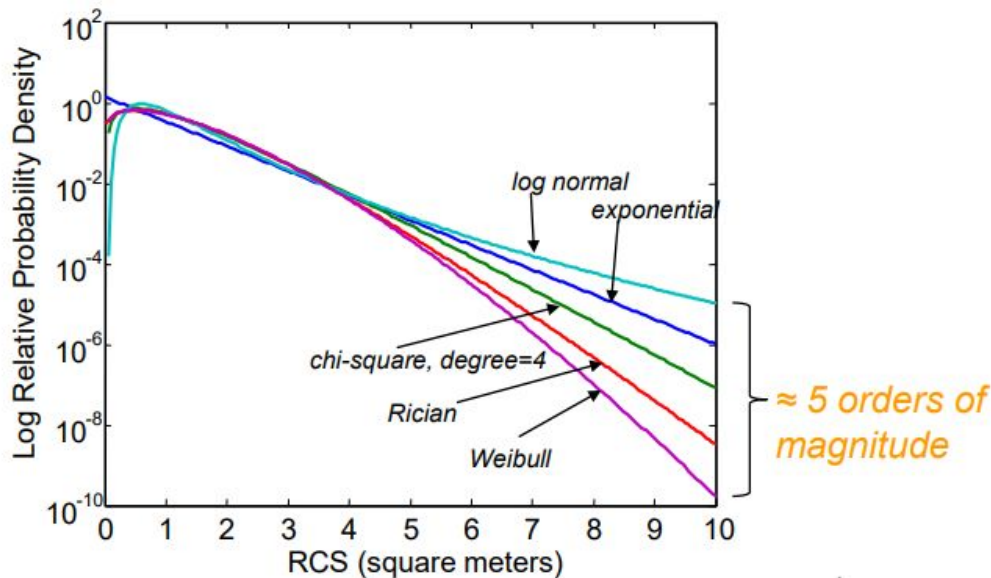
Comparison of RCS PDFs - 1

- **Variance = 0.5 in all cases**
- **Mean = 1 for all except exponential**
 - Exponential mean = 0.707



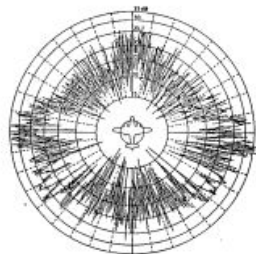
Comparison of RCS PDFs - 2

- Same PDFs, but now on a log scale
- Major differences in PDF tails
 - Note the long “tail” of the log-normal distribution in this example
- A higher or lower “tail” implies a significantly higher or lower frequency of large-amplitude RCS “spikes”
 - Large RCS values still rare



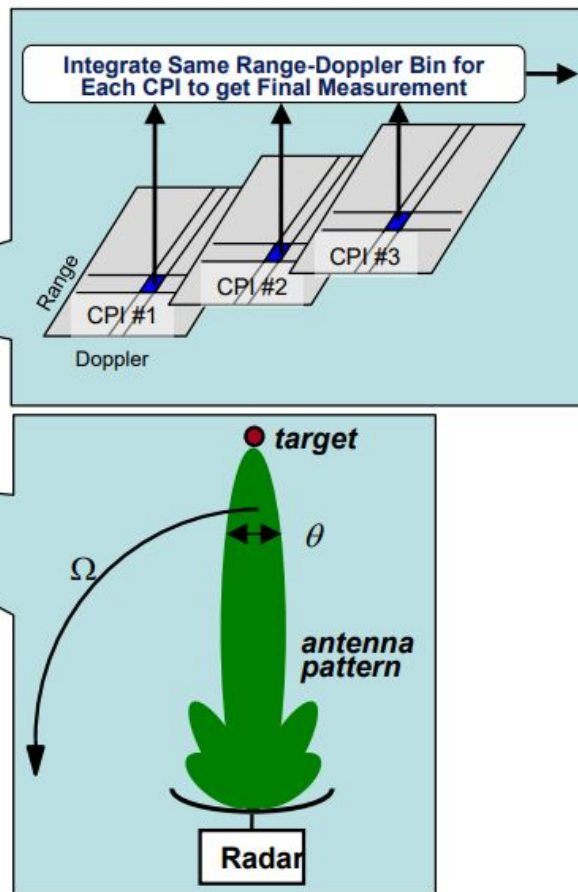
Decorrelation in Aspect and Frequency

- RCS varies with both frequency and aspect angle. How much change in RF or angle is required to significantly change (“decorrelate”) the RCS value?
- **The decorrelation interval in aspect angle is** $\Delta\theta \approx \frac{c}{2L_w F}$
 - L_w is interpreted as the target size **perpendicular** to the line of sight (“width” as viewed from radar)
- **The decorrelation interval in frequency is** $\Delta F \approx \frac{c}{2L_d}$
 - L_d is interpreted as the target size **along** the line of sight (“depth” as viewed from radar)
 - *We'll return to this later in context of frequency agility*
- **Example: 5 m long target**
 - $\Delta\theta = 1.7 \text{ deg @ } 1 \text{ GHz}, 0.018 \text{ deg @ } 95 \text{ GHz}$
 - $\Delta F = 30 \text{ MHz}$
- **But why do we care?**



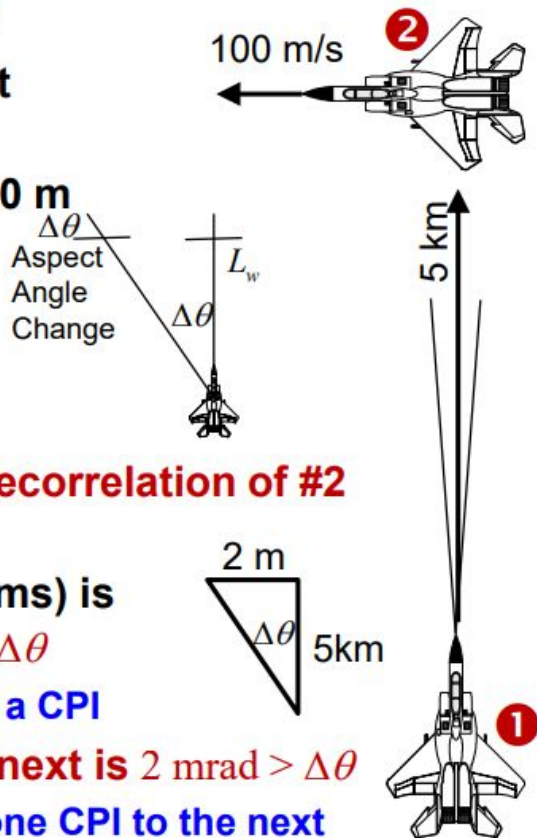
Detection Using N Measurements

- A single threshold test is often based on **noncoherent** integration of N repeated “**measurements**”, instead of just 1
- N measurements may be collected in various ways
 - Examining the same range-Doppler bin from each of N CPIs in a dwell
 - “Measurement” $\equiv$ range-Doppler bin value
 - New measurement value with each CPI
 - Examining the same range bin for N successive pulses
 - For instance, with a rotating antenna
 - “Measurement” $\equiv$ range bin value
 - New measurement value with each pulse
- If the data happens to contain a target, how are the statistics of those N target measurements related?
- The answers turn out to make a significant difference to detection performance prediction



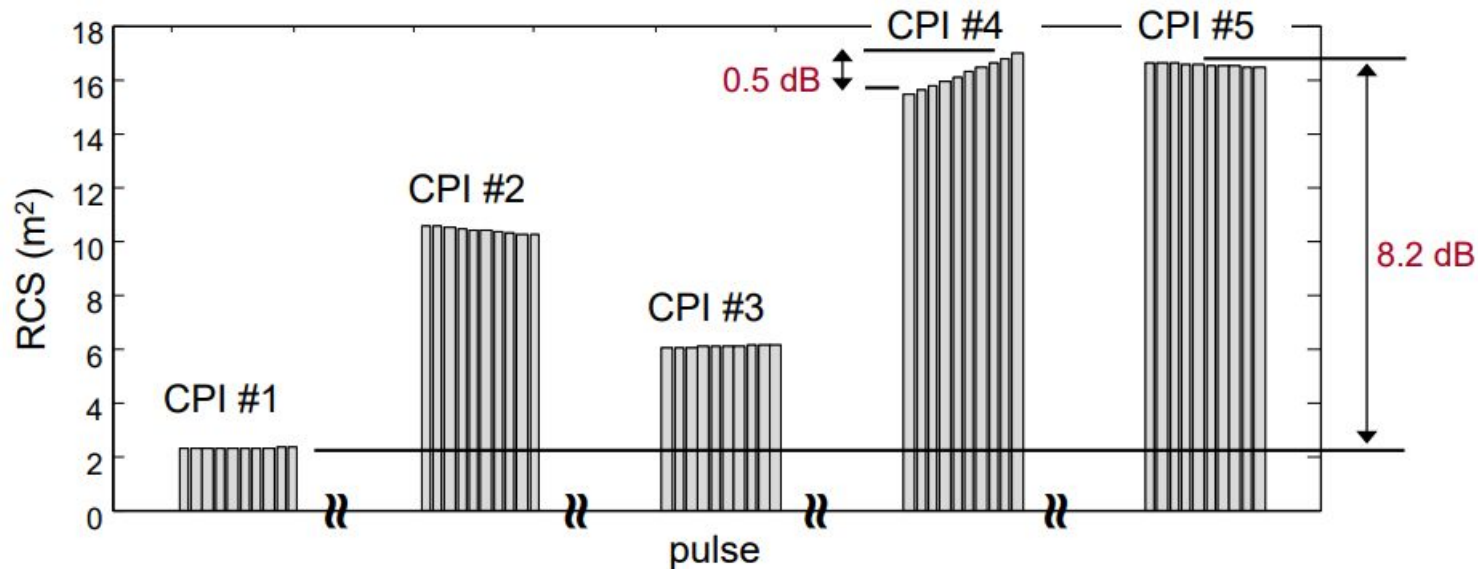
RCS Correlation Dynamics Example - 1

- Aircraft #1 views crossing aircraft #2
- Aircraft #2 traveling at 100 m/s (about 224 mph) at nominal range of 5 km
- The width of #2 as viewed from #1 ≈ 10 m
 - About the size of an F-16
- 10 GHz radar on #1
 - CPI: 20 pulses @ 1 kHz PRF = 20 ms
 - 100 ms between CPIs
- The angular change $\Delta\theta$ needed for decorrelation of #2 RCS is $c/2L_wF = 1.5$ mrad
- The angular change within 1 CPI (20 ms) is $(0.02 \text{ sec})(100 \text{ m/s})/(5000 \text{ m}) = 0.4 \text{ mrad} < \Delta\theta$
 - So we expect no decorrelation within a CPI
- The angular change from one CPI to next is $2 \text{ mrad} > \Delta\theta$
 - So we do expect decorrelation from one CPI to the next
- Let's see if it works out that way ...



RCS Decorrelation Dynamics Example - 2

- Model aircraft #2 as 20 randomly-located, unresolved scatterers in a 5 m × 5 m moving box
- The observed RCS of the resulting simulated measurements:



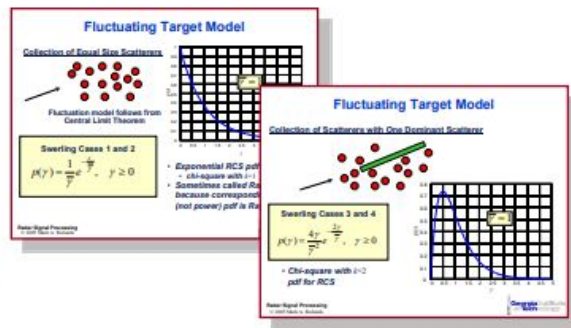
- The RCS is approximately constant within a CPI, but ...
- We see significant RCS variations from one CPI to the next

Some Decorrelation Terminology

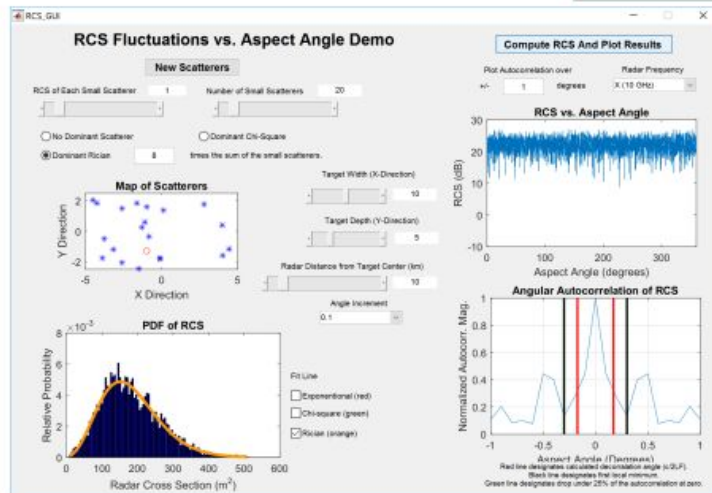
- Our main concern is the correlation between N measurements that are combined to perform one threshold test
 - E.g., the same range bin on successive pulses, or the same range-Doppler bin on successive CPIs
 - We usually assume the data is uncorrelated from one threshold test to the next → statistically independent threshold tests
- “Slow decorrelation” is when all N measurements have approximately the **same** RCS (power) drawn from the RCS PDF
 - Implies approximately constant aspect angle and RF during the time to collect those N measurements (or a nonfluctuating target)
- “Fast decorrelation” is when each of the N measurements has a **different** and independent value of RCS (power) drawn from the RCS PDF
 - Implies significant changes in aspect angle and/or RF from one measurement to the next
- Reality is often between these two bounding extremes

MATLAB GUI RCS: RCS of Complex Targets

- Simulates the angular variation of the RCS of a “complex target” composed of “many” scatterers
 - With or without “a dominant scatterer”
- Reproduces this experiment: 
 - Can vary # of scatterers, box (“target”) size, the presence or absence of a dominant scatterer and its relative strength
 - Compare to various PDFs
 - Compute autocorrelation lengths



- >> RCS
 - Select/change target characteristics, then click on ‘New Scatterers’
 - Select/change plot options, then click on ‘Compute RCS and Plot Results’



Swerling Models for Target RCS - 1

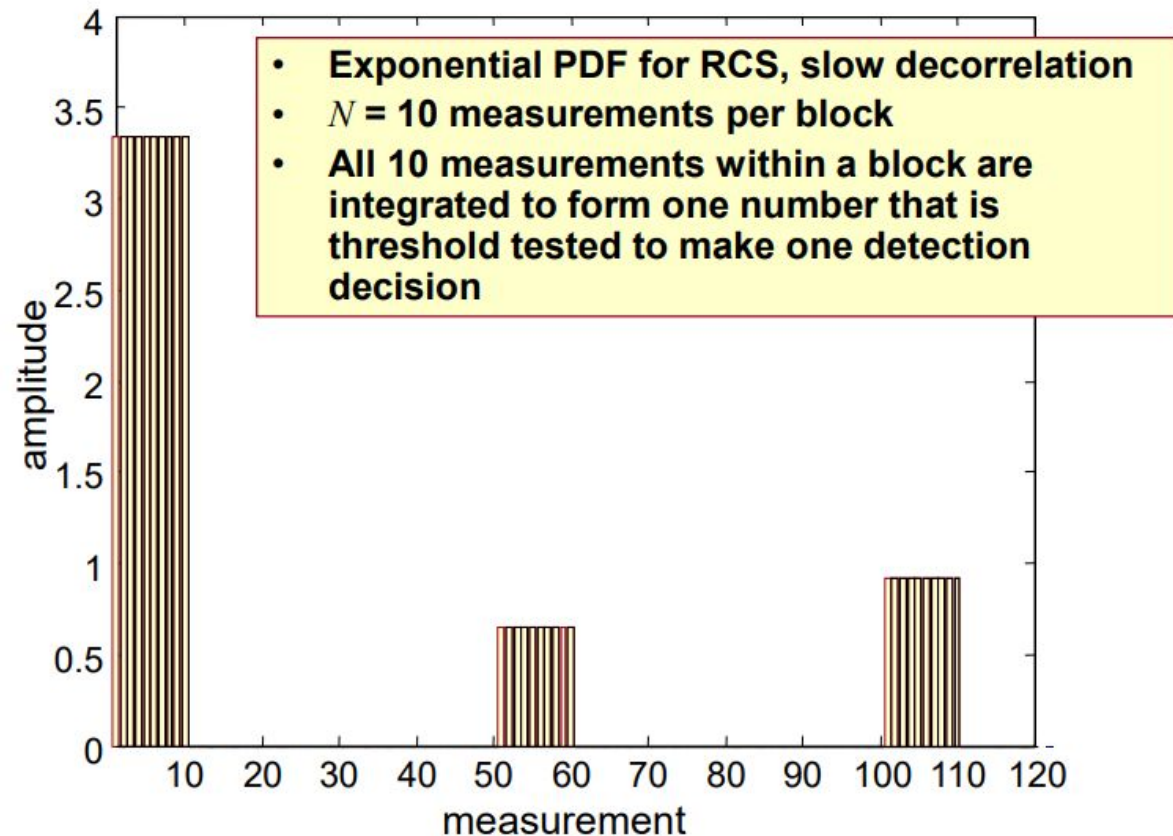
- A Swerling model is an assumption about the statistics of a block of N measurements of target power (RCS) that will be noncoherently integrated to perform one threshold test
- The purpose of the model is to allow a mathematical prediction of probability of detection
- A traditional Swerling model chooses one each of
 - Two RCS PDFs: exponential or 4th-degree chi-square
 - Two decorrelation extremes: fast or slow
- The nonfluctuating case is sometimes called the “Swerling 0” or “Swerling 5” case

Swerling Models for Target RCS - 2

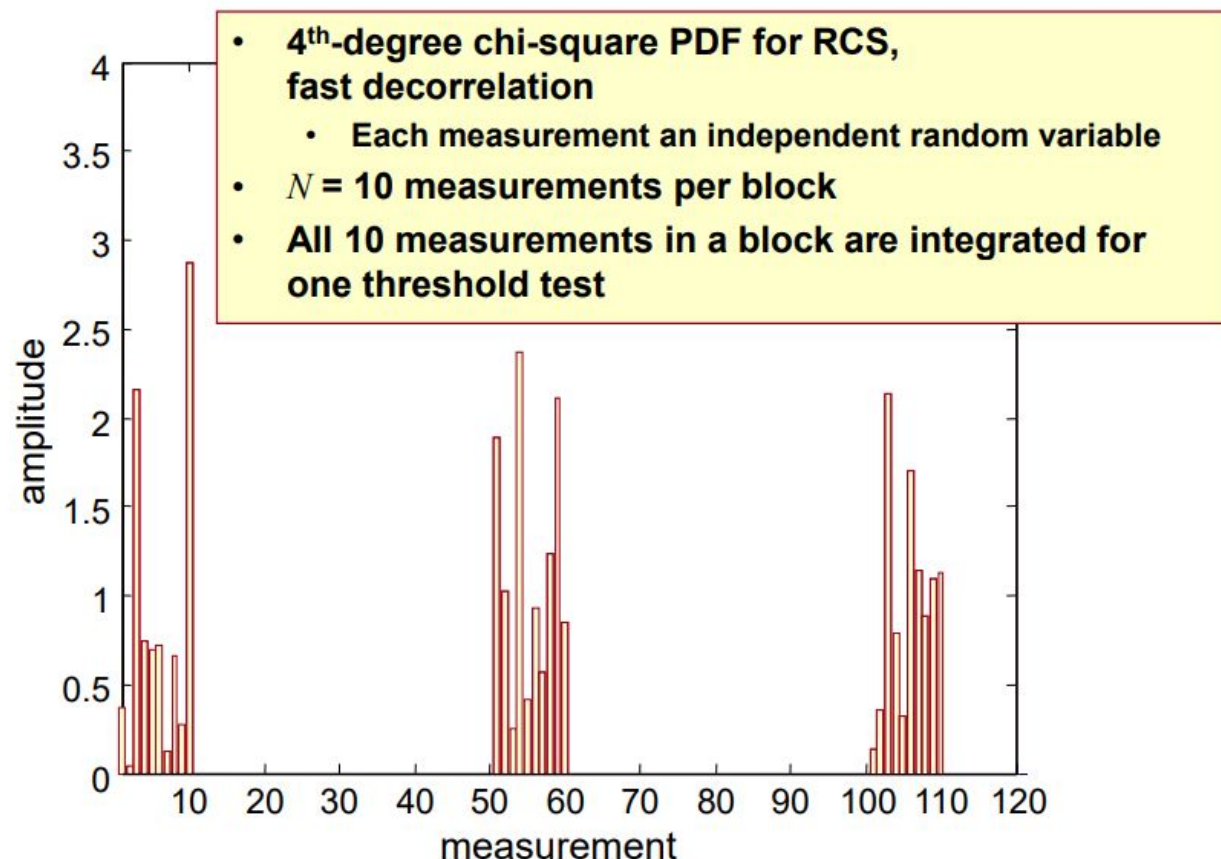
<i>Probability Density Function of RCS</i>	<i>Decorrelation</i>	
	<i>Slow</i>	<i>Fast</i>
exponential	Case 1	Case 2
chi-square, degree 4	Case 3	Case 4

- **Swerling models are important because**
 - They represent reasonable bounding cases for many moderate- and low-resolution radar scenarios
 - A body of detection theory has been built up based on them
 - They represent a generalizable strategy for making detection probability predictions
 - We can pair other PDFs with a decorrelation model to get a non-Swerling model

Three Blocks of Swerling 1 Data for Three Threshold Tests



Three Blocks of Swerling 4 Data for Three Threshold Tests



Statistical Models for Clutter

We Need Statistical Models of Clutter, Too

- Recall that for “large” N and completely random relative phases, the expected value of the power or RCS is proportional to the sum of the individual power or RCS values
 - Random phases implies randomly distributed locations within the cell

$$\bar{P}_r, \bar{\sigma} \propto \sum_{i=1}^N \sigma_i$$

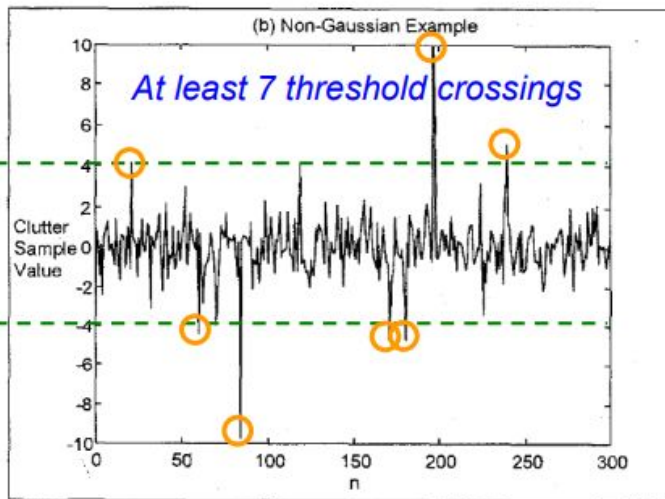
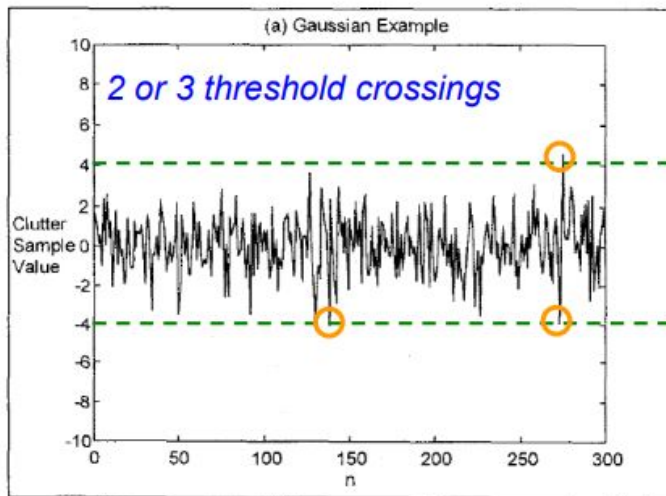
- Scatterers combine on a “power basis”
 - Most relevant to scattering for surface clutter, weather, chaff, etc.
- Just as with targets, however, accurate detection calculations require characterization of the statistical variation of the clutter samples that enter into a detection test
- So we also need to model clutter PDFs and decorrelation properties ...

Clutter Reflectivity PDFs

- Clutter reflectivity varies statistically with time **within** a resolution cell due to scatterer motion (e.g., wind, wave motion), radar-target motion, etc.
- Clutter reflectivity varies statistically **between** range/angle/cross-range resolution cells (i.e., spatially) due to terrain variations
- Many of the same PDFs are used to model clutter statistics
 - At coarser resolution, exponential
 - At finer resolution, more use of long-tailed distributions such as K, log-normal, Weibull
 - “Compound distributions” combine one PDF for cell-to-cell variation, another for variation with time
- **Longer-tailed clutter interference PDFs tend to force higher thresholds for a given P_{FA} $\rightarrow$ higher signal-to-interference ratios (SIRs) are needed to achieve a given P_D**

Example of “Spikey” Clutter

- Gaussian I and Q → Rayleigh voltage, exponential power
- Non-Gaussian I and Q
 - Likely models are Weibull or K PDF



A. D. Keckler, D. L. Studeman, and D. D. Weiner, "Non-Gaussian Clutter Modeling and Application to Radar Target Detection", *Proc. IEEE National Radar conference*, pp. 240-245, 1997.

• Spikey clutter requires a higher threshold for a given P_{FA}

Clutter Decorrelation Models

- Decorrelation models imply a Doppler power spectrum $S(F_D)$ associated with a series of reflectivity measurements
 - Fast decorrelation $\rightarrow$ white spectrum (wide)
 - Slow decorrelation $\rightarrow$ asinc spectrum (narrow)

- Clutter decorrelation is usually characterized by a power spectrum

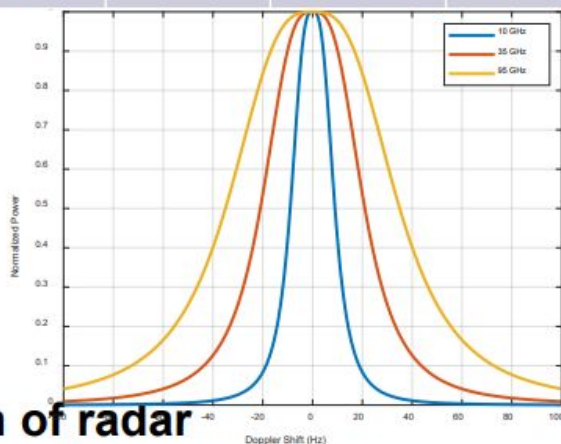
Clutter	10 GHz	35 GHz	95 GHz
Trees, 6-25 mph wind	$F_c = 9$	$F_c = 21$	$F_c = 35$

- Some traditional power spectrum models proposed for clutter:

- Cubic: $S(F_D) = \frac{A}{1 + (|F|/F_c)^3}$

- Gaussian: $S(F_D) = Ae^{-\alpha(F/\sigma_F)^2}$

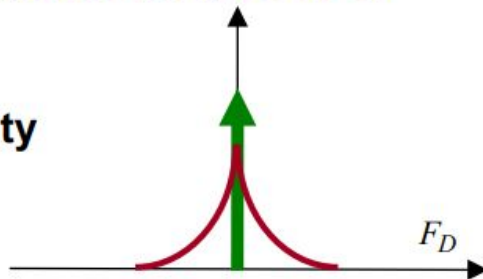
- The model parameters are a function of radar characteristics and environmental conditions



Another Example: Billingsley Clutter Decorrelation Model

- A relatively recent model popular in the space-time adaptive processing community combines a “**DC**” and an “**AC**” term

$$P(F_D) = \frac{\alpha}{\alpha+1} \delta_D(F_D) + \frac{1}{\alpha+1} \cdot \frac{\lambda\beta}{4} \exp\left(-\frac{\lambda\beta}{2}|F_D|\right)$$



- F_D = Doppler frequency, δ_D = Dirac impulse function
- The DC/AC ratio α depends on the type of clutter, windspeed w in mph, and F_0 (GHz):

$$\alpha = k_{ct} 489.9 \cdot w^{-1.55} F_0^{-1.21}$$

– k_{ct} = weighting factor specific to clutter type

$w \equiv \text{miles/hour}$
 $F_0 \equiv \text{GHz}$
- The slope parameter β depends on windspeed:

$$\beta^{-1} = 0.1048 [\log_{10} w + 0.4147] \quad \beta \equiv \text{meters/sec}$$

Statistical Model of Receiver Noise

Noise Signal Statistics

- Time domain behavior:
 - The I and Q receiver output noise voltages are independent, identically-distributed (iid) zero-mean Gaussian random processes with identical variances
- It follows that ...
 - The magnitude of the complex signal $I+jQ$ is Rayleigh distributed ...
 - ... and the squared-magnitude is exponential-distributed ...
 - ... and the phase angle $\tan^{-1}(Q/I)$ is uniformly distributed on $[0, 2\pi)$
 - Proof of all of these is a common example of “one function of two random variables” in many random processes textbooks
- Frequency domain behavior: recall that the power spectrum is approximately white:

$$S_n(F) \approx kT \text{ W/Hz (a constant)}$$